Edge Speech Recognition and Wireless Cloud Telemetry on STM32

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Introduction and Motivation

An Edge AI system for hands-free industrial control. It processes audio on-device and transmits only actionable inferences (e.g., "Yes/No") to the cloud via cellular, ensuring privacy and saving bandwidth.

- **Problems Addressed:** High bandwidth costs, transmission latency, high power consumption, and privacy risks associated with continuous cloud audio streaming.
- **The Solution**: STM32F407 MCU combined with Machine Learning and FreeRTOS multitasking architecture.

Layered Hardware Architecture

- **Main MCU:** STM32F407 Discovery (ARM Cortex-M4F, 112KB RAM, FPU).
- **Sensor:** On-board MEMS Microphone. Generates PDM (Pulse Density Modulation) signals, captured seamlessly into memory via I2S hardware and DMA (Direct Memory Access) without stressing the CPU.
- **Telemetry Module:** Sixfab 3G/4G/LTE Base Hat (Quectel Modem).



Fig. 1: Sixfab 3G/4G/LTE Base HAT Hardware Module.

- **Layered Architecture:** Hardware dependencies are abstracted using custom driver layers (`drv_microphone`, `drv_atcommand`). This allows the application tasks to communicate seamlessly through upper middleware layers (`Cellular`), ensuring modularity and code portability.

RTOS & Concurrency

- **Tasks:**
  - *Audio Task (Acquisition/Audio Preprocessing/AI)*
  - *Network Task*
- **Inter-Process Communication:** Message Queues route audio streams and inference strings safely between tasks.
- **Resource Management:** Mutexes and Semaphores eliminate **Race Conditions** on the shared UART communication line.

Non-Functional Requirements

- **Latency (Performance):** Eliminates cloud round-trip times by executing DSP and neural network inferences purely at the edge.
- **Privacy (Security):** Ensures strict data confidentiality by transmitting only binary classification states, preventing raw audio interception.
- **Resource Efficiency:** Architected to operate reliably within strict constraints (192KB total SRAM, low-power Cortex-M4F capabilities).

System Structure

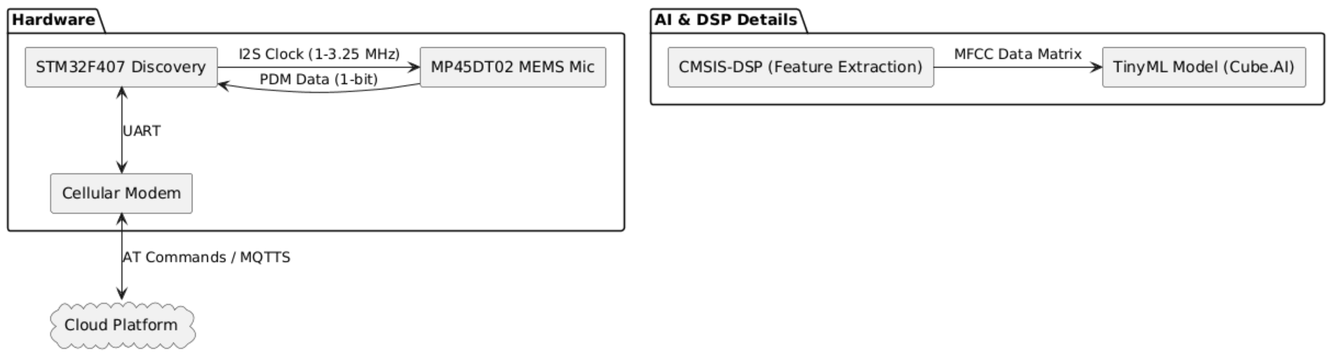


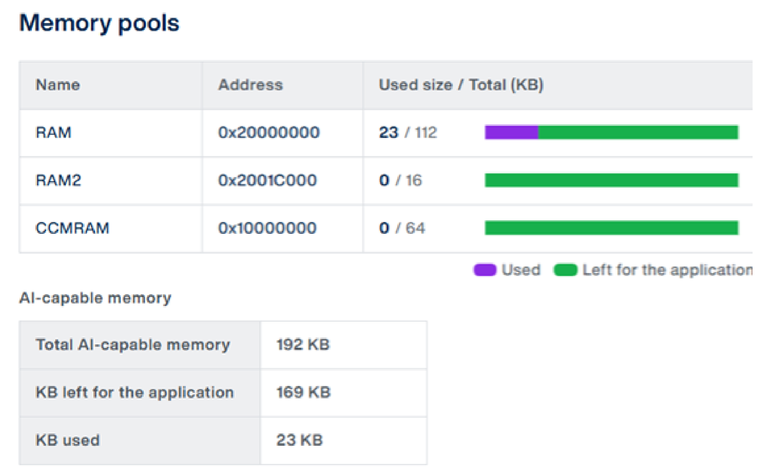
Fig. 2: System Architecture Block Diagram (PlantUML)



Fig. 3: STM32CubeMX Middleware and Software Packs Configuration

TINYML & Memory Optimization

- **DSP Pipeline:** Raw PDM → PCM conversion → STM32 AI Audio Preprocessing Library (Windowing, DCT, Mel-Filterbank) → Mel-Spectrogram.
- **Model Inference:** Quantized KWS Keras model optimized via STM32Cube.AI.
- **Memory Management: Static Memory Allocation** (`.bss`/`.data`) to guarantee real-time safety and eliminate dynamic heap overhead within the 112KB RAM limit.



Memory pools

Name	Address	Used size / Total (KB)
RAM	0x20000000	23 / 112
RAM2	0x2001C000	0 / 16
CCMRAM	0x10000000	0 / 64

Used · Left for the application

AI-capable memory

Total AI-capable memory	192 KB
KB left for the application	169 KB
KB used	23 KB

Fig. 4: STM32Cube.AI Memory Pool Utilization Report

Unit Testing & Validation

- **Hardware-Independent Framework:** Utilized the CppUTest framework to validate software components locally via rigorous Hardware Abstraction Layer (HAL) mocking.
- **Modular Verification:** Isolated test suites independently prove the stability of I2S DMA callbacks (`test\_microphone`), Cellular AT command parsing (`test\_cellular`), and MQTT telemetry formatting (`test\_mqtt`).
- **Pipeline Integrity:** Automated tests dynamically guarantee the accuracy of DSP feature extraction (`test\_audio\_dsp`) and FreeRTOS task transitions (`test\_ai\_app`) prior to edge deployment.

Security & Product Readiness

- By extracting insights at the edge and only sending the integers (1/0) via MQTT, the system implements **Privacy by Design**.

Results & Future Work

- **Achievements:** Successfully implemented a multi-layered software architecture. The AI inference engine (STM32Cube.AI) and the custom-built **Cellular AT-Command Middleware** are now fully integrated under FreeRTOS, compiling with zero errors and an optimized memory footprint. Telemetry data (1/0) is transmitted securely via MQTT.

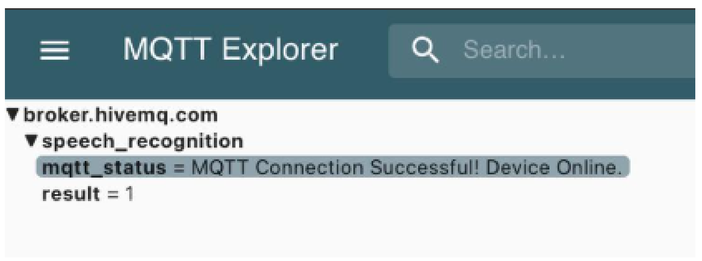


Fig. 5: MQTT broker log validating real-time wireless transmission of edge inference strings over the cellular network

- **Future Work:** Evaluating model accuracy under varying background noise (SNR) levels and conducting full system power simulations using *Model-Based Design* approaches.

Fig. 6: Scan to See the Entire Project Repository and Technical References